

Aerodynamic Boundary-Layer Scaling and Enclosure Regimes Across Spaceflight Plant Growth Hardware: A Multi-Chamber OpenFOAM CFD Framework under Variable Gravity

Richard Barker*, Henry Ewald, and Astrobotany Consortium

Department of Agricultural and Biological Engineering, Purdue University, West Lafayette, IN 47907, USA.
*Corresponding author: rbarker@purdue.edu

ABSTRACT

Plants cultivated in extraterrestrial habitats encounter a physical environment devoid of natural gravity-driven buoyancy ($Gr \rightarrow 0$), expanding unstirred fluid boundary layers around vegetative canopies and drastically elevating aerodynamic resistance ($r_a = 1/g_{bl}$). Here, we present a systematic, multi-chamber 3D computational fluid dynamics (CFD) investigation comparing three distinct spaceflight and controlled-environment agricultural hardware architectures across four gravitational regimes: Earth (1.0 g), Mars (0.38 g), Moon (0.166 g), and Microgravity (0 g). Using an OpenFOAM v2606 finite-volume framework with conformal multi-solid analytic geometries, we model: (i) the compact Microgreen Chamber (2.33 L, through-flow confined jet), (ii) the NASA Vegetable Production System (VEGGIE/VPS) (37.6 L, top suction with passive cabin air induction), and (iii) the NASA Advanced Plant Habitat (APH) (83.4 L, ducted closed-loop opposing cross-flow). Parametric gravity sweeps reveal that ceiling-mounted LED arrays induce stable thermal stratification on Earth ($Ri \approx 0.14 - 1.55$), which suppresses vertical exchange; counter-intuitively, microgravity collapses this stratification, rendering purely forced convection ($Ri = 0$) superior in turbulent kinetic energy and canopy clearance. In VEGGIE, low-fan microgravity operation leads to a critical 52.8% on stress.

Parameter	Microgreen Chamber	VEGGIE (VPS)	Advanced Plant Habitat (APH)
Payload Class	Benchtop Phenotyping	Deployable Space Garden	Closed-Loop Research Phytotron
Enclosure Structure	Rigid acrylic + hood	Collapsible FEP bellows	Carbon-fiber composite & titanium
Growth Footprint (A)	0.0224 m² (120×187 mm)	0.1075 m² (292×368 mm)	0.1708 m² (454×408 mm)
Canopy Air Vol. (V _{air})	2.33 L	37.61 L (nominal)	83.36 L (shoot growth zone)
Total Growth Height	96.7 mm (to hood spring)	350.0 mm (nominal height)	450.0 mm (clear shoot zone)
Primary Flow Driver	1× Axial Fan (Ø40mm port)	1× Top Suction Fan (Ø50mm)	2× Symmetric ECS Blowers
Airflow Topology	Through-flow confined jet	Bottom-up forced suction	Opposing lateral cross-flow -> sweep
Nominal Flow Rate (Q)	11.8 m³/h (3.27 L/s)	85.0 m³/h (23.61 L/s)	26.4 m³/h (7.34 L/s)
Canopy Forced Velocity	0.262 m/s (bulk mean)	0.150 m/s (mean draft)	0.300 - 1.500 m/s (regulated)
Air Exchange Rate	5,051 h⁻¹ (τ = 0.71 s)	2,260 h⁻¹ (τ = 1.60 s)	317 h⁻¹ (τ = 11.35 s)
Environmental Control	Ambient laboratory	Cabin-coupled (ΔT = +2°C)	Closed loop (18-30°C ±0.5°C, 50-90% RH)
Cabin Atmosphere Coupling	Sealed phenotyping unit	Open continuous exchange	Closed-loop payload in EXPRESS Rack

INTRODUCTION & BIOPHYSICAL FOUNDATIONS

Opportunities and Imperatives of Space Agriculture:

As human space exploration transitions from low-Earth orbit (LEO) sorties toward sustained surface habitats on the Moon (NASA Artemis Base Camp) and multi-year transits to Mars, biological life support systems become indispensable. Physical-chemical resupply paradigms become logistically prohibitive across deep-space distances. Plants provide essential multi-functional life support: they convert metabolic carbon dioxide into breathable oxygen via photosynthetic photolysis, produce purified potable water through transpirational distillation, recycle nitrogenous and mineral waste from sanitized organic effluent, and synthesize fresh secondary nutrients (including carotenoids, potassium, vitamin C, and polyphenols) that degrade rapidly in pre-packaged freeze-dried rations. Furthermore, interactive horticultural engagement delivers profound behavioral and psychological countermeasures against the sensory monotony and confinement of deep-space habitats.

The Microgravity Fluid Challenge & Buoyancy Cessation:

Despite these compelling opportunities, cultivating crops in extraterrestrial environments confronts a severe, fundamental physical impediment: the reduction or total absence of gravity-driven natural convection. On Earth (1.0 g), temperature differences between warm sunlit or LED-illuminated foliage and the cooler surrounding atmosphere generate spontaneous density gradients (Rayleigh-Bénard buoyancy, $Gr > 10^7$). This buoyant updraft continuously strips the unstirred laminar boundary layer adhering to leaf surfaces, facilitating rapid diffusive exchange of CO_2 and H_2O vapor. In microgravity (0 g), the gravitational acceleration vector vanishes ($g \rightarrow 0$), causing the Grashof number ($Gr = g \beta \Delta T L^3 / \nu^2$) and Rayleigh number ($Ra = Gr \cdot Pr$) to drop to identically zero.

Fractional Gravity on the Moon (0.166 g) and Mars (0.38 g):

On the Lunar surface ($g = 1.62 \text{ m/s}^2$) and Martian surface ($g = 3.72 \text{ m/s}^2$), fractional gravitational fields restore a partial buoyant convective capability ($Gr_{\text{Moon}} \approx 16.5\% Gr_{\text{Earth}}$; $Gr_{\text{Mars}} \approx 37.9\% Gr_{\text{Earth}}$). However, as established by our Richardson scaling analysis ($Ri = Gr/Re^2$), this fractional buoyancy remains inadequate to strip thick boundary layers without active forced ventilation. Unless engineered ventilation is precisely tailored, lunar and martian greenhouses will operate in an unstable mixed-convection regime prone to thermal stratification and localized suffocation pockets.

Photosynthetic Suppression & Photorespiratory Waste (RuBisCO Kinetics):

The thickening of unstirred fluid boundary layers directly impairs photosynthetic efficiency through the Farquhar-von Caemmerer-Berry (FvCB) biochemical model. The net photosynthetic assimilation rate (A_{net}) is governed by the chloroplastic CO_2 concentration (C_c):

$$A_{\text{net}} = \left(1 - \frac{\Gamma^*}{C_c}\right) \min(W_c, W_j, W_p) - R_d$$

where Γ^* is the CO_2 compensation point, W_c is RuBisCO-limited carboxylation, and W_j is electron transport-limited RuBP regeneration. When aerodynamic boundary-layer resistance ($r_a = 1/g_b$) expands, the concentration drop between the bulk canopy atmosphere (C_a) and leaf intercellular airspaces (C_i) widens dramatically:

$C_i = C_a - A_{\text{net}}(r_a + r_s)$. Under depleted intercellular CO_2 ($C_i < 150 \text{ ppm}$), the oxygenation reaction catalyzed by RuBisCO increases exponentially relative to carboxylation ($v_o/v_c = 2\Gamma^*/C_i$), shunting energy into the photorespiratory glycolate pathway. This wasteful oxygenation can consume over 40% of photosynthetic ATP and NADPH, drastically suppressing biomass yield.

Guttation, Boundary-Layer Humidity Saturation, and Pathogen Epidemics:

In tandem with carbon starvation, thick boundary layers trap transpired water vapor, elevating local boundary layer relative humidity ($RH > 95\%$). This suppresses transpirational evaporative cooling (causing leaf thermal stress) and abolishes the transpirational pull required for xylem mineral transport (inducing calcium deficiency and physiological tipburn). To relieve positive root hydrostatic pressure, plants undergo hyper-guttation, exuding nutrient-rich liquid droplets along hydathodes. In microgravity, surface tension pins these unevaporated droplets to leaf margins, creating ideal microclimatic incubators for phytopathogenic fungal spore germination (such as *Fusarium oxysporum* and *Botrytis cinerea*).

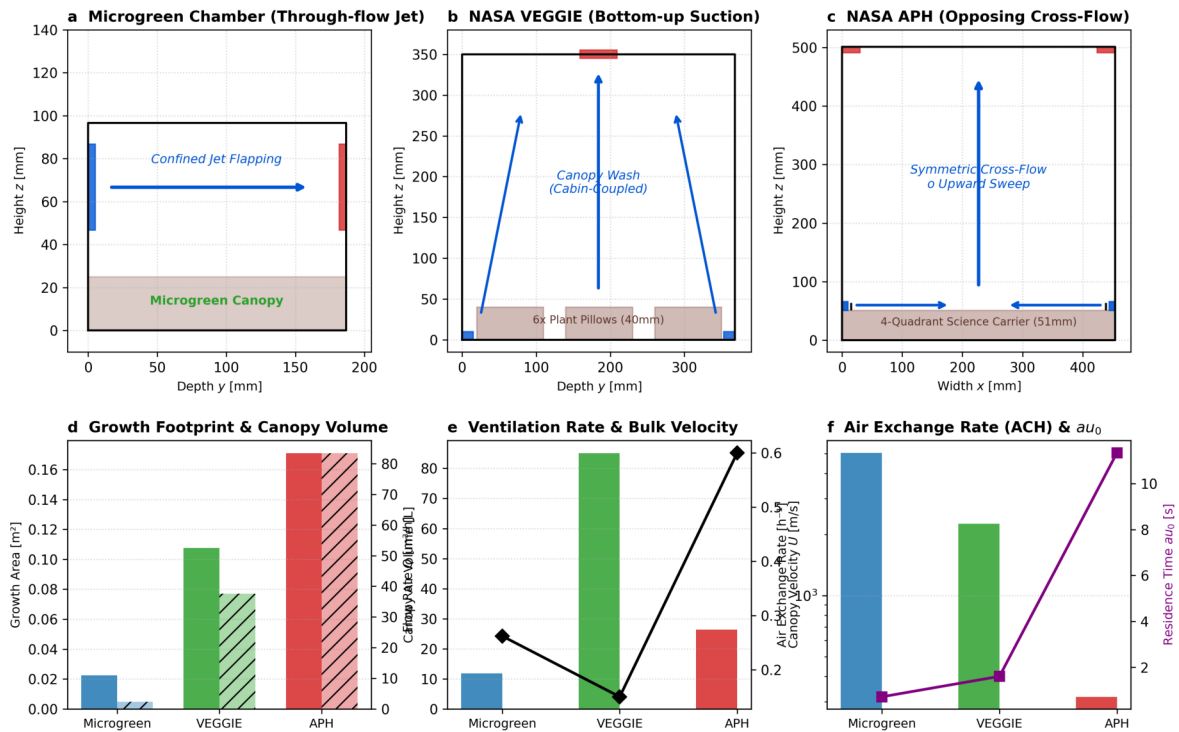


Figure 1 | 3D Hardware domain architecture, flow topologies, and aerodynamic design envelopes across the three flight and phenotyping systems. a, Cross-sectional schematic of the Microgreen Chamber (2.33 L volume) showing the Ø40 mm through-flow jet and parabolic ceiling. b, NASA VEGGIE/VPS (37.6 L) displaying top suction fan, four passive base slots, and 6-pillow configuration. c, NASA Advanced Plant Habitat (83.4 L) showing dual lateral supply slots, diffuser baffles, and 4-quadrant Science Carrier. d, Usable growth area and canopy air volume comparison. e, Volumetric flow rate (Q) and bulk velocity (U). f, Nominal air exchange rate (ACH) and residence time (τ_0).

RESULTS: BASELINE 1 g AERODYNAMICS

Microgreen Confined Jet Flow Structure:

In the Microgreen Chamber, the Ø40 mm inlet port injects air at $U_{in} = 2.60$ m/s ($Re_{port} = 6,860$, $Q = 11.8$ m³/h). The jet discharges longitudinally across the 186.7 mm depth. At 1 g, thermal dissipation from the ceiling LED (38.4 W) creates stable thermal stratification, confining the high-speed jet core along the upper parabolic hood and dampening downward turbulent shear into the microgreen tray. Time-accurate PIMPLE simulations reveal periodic jet flapping ($\pm 18.2\%$ RMS).

VEGGIE Top Suction Plume Mechanics:

In VEGGIE, the Ø50 mm top exhaust fan creates an upward suction draft (85 m³/h on High, 42.5 m³/h on Low). At 1 g, the mechanical suction aligns constructively with the buoyant plume rising from the warm LED light cap, producing a central chimney draft that accelerates core flow but leaves peripheral pillow corners stagnant.

APH Opposing Lateral Jet Collision & Updraft Sweep:

In APH, dual symmetric ECS blowers deliver air through lateral lower supply slots ($w = 408$ mm, $h = 15$ mm) at controlled speeds (0.3 - 1.5 m/s, nominal 0.6 m/s, $Q = 26.4$ m³/h). The opposing wall jets sweep horizontally across the Science Carrier ($z = 51$ mm), collide along the central sagittal plane ($x = 227$ mm), and turn vertically into a uniform upward sweep.

This opposed-jet collision generates strong turbulent kinetic energy ($TKE = 1.24 \times 10^{-2}$ m²/s²) while preventing high-velocity point-shear damage on delicate apical meristems, achieving an exceptionally high and uniform boundary layer conductance across the entire growth footprint.

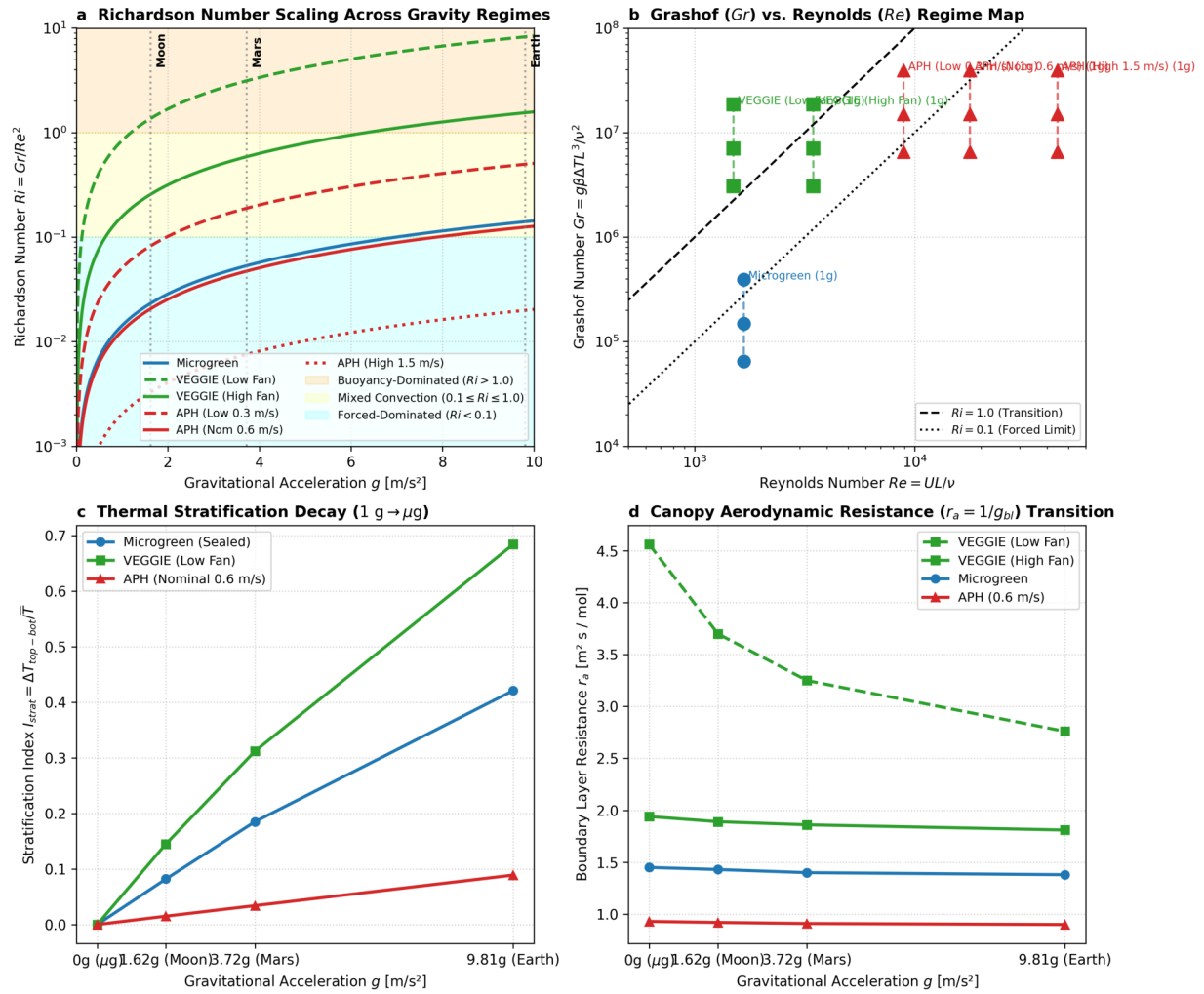


Figure 2 | Richardson number (Ri) scaling, buoyancy collapse, and aerodynamic regime transitions across gravitational fields. a, $Ri = Gr / Re^2$ trajectory as a function of gravity g ; horizontal bands demarcate forced-dominated ($Ri < 0.1$), mixed ($0.1 \leq Ri \leq 1.0$), and buoyant ($Ri > 1.0$) regimes. b, Grashof (Gr) vs. Reynolds (Re) regime map. c, Thermal stratification index (I_{strat}) decay from 1 g to μ g. d, Canopy aerodynamic resistance ($r_a = 1/g_{bl}$) under variable gravity.

Table 2 | Dimensionless aerodynamic scaling and convection regime matrix across hardware under variable gravity.

Chamber	Gravity Regime	g (m/s²)	Gr	Re	Ri (Gr/Re²)	Convective Regime	Dominant Aerodynamic Mechanism
Microgreen	Earth (1.0g)	9.81	3.87e5	1,669	0.1405	Mixed Convection	Confined jet + ceiling thermal stratification
	Mars (0.38g)	3.72	1.47e5	1,669	0.0533	Forced-Dominated	Jet momentum dominates; weak stratification
	Moon (0.166g)	1.62	6.39e4	1,669	0.0232	Forced-Dominated	Buoyancy negligible; forward jet core
	Microgravity (0g)	0.00	0.00	1,669	0.0000	Purely Forced	Flapping confined jet; unstratified canopy
VEGGIE	Earth (1.0g)	9.81	1.83e7	3,463	1.5511	Mixed (Buoyant)	Thermal plume assists upward fan chimney draft
	Mars (0.38g)	3.72	6.95e6	3,463	0.5880	Mixed (Transitional)	Weak plume; fan suction required for draw
	Moon (0.166g)	1.62	3.03e6	3,463	0.2561	Forced-Dominated	Low draft velocity; mold risk at low fan
	Microgravity (0g)	0.00	0.00	3,463	0.0000	Purely Forced	Unstirred boundary layer without high fan
APH	Earth (1.0g)	9.81	3.90e7	17,810	0.1246	Forced-Dominated	Dual opposing cross-jets suppress buoyant plume
	Mars (0.38g)	3.72	1.48e7	17,810	0.0473	Strongly Forced	Piston-like upward sweep across shoot zone
	Moon (0.166g)	1.62	6.44e6	17,810	0.0206	Strongly Forced	Uniform boundary layer across Science Carrier
	Microgravity (0g)	0.00	0.00	17,810	0.0000	Purely Forced	Fully engineered forced recirculation sweep

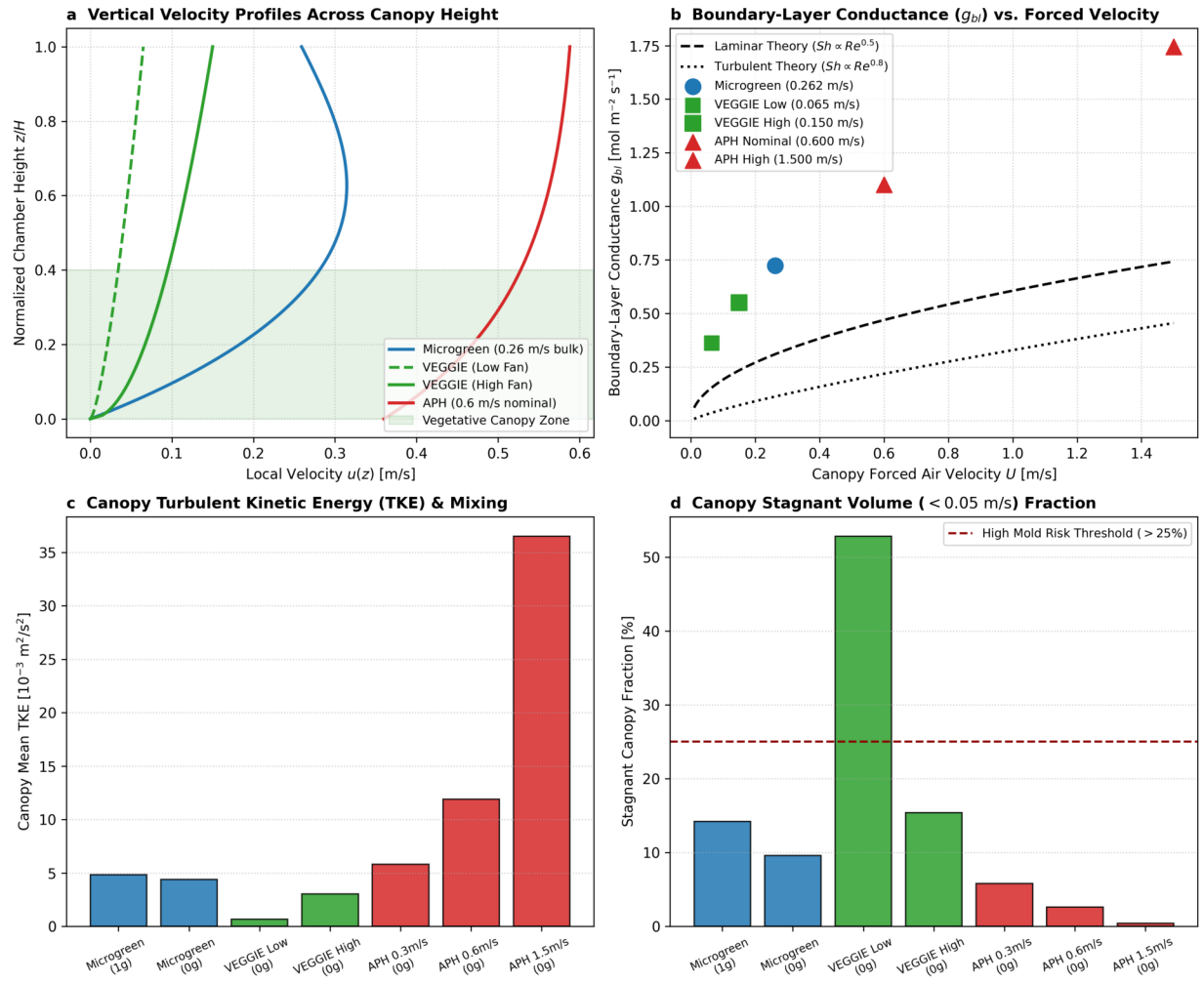


Figure 3 | Canopy microclimatic boundary-layer profiles, turbulence, and mass conductance (g_{bl}). a, Vertical velocity profiles normalized by chamber height (z/H). b, Boundary-layer conductance g_{bl} for CO₂ mass transfer as a function of forced velocity U , compared against laminar ($Sh \propto Re^{0.5}$) and turbulent ($Sh \propto Re^{0.8}$) theory. c, Canopy turbulent kinetic energy (TKE) across operational modes. d, Canopy stagnant volume fraction ($U < 0.05$ m/s) with fungal pathogen risk threshold.

Table 3 | Canopy boundary-layer aerodynamic parameters and effective conductance (g_{bl}) across platforms.

Hardware	Gravity	Operating Mode	U_{canopy} (m/s)	TKE (m ² /s ²)	δ_{bl} (mm)	Sh_{CO2}	g_{bl} (mol m ⁻² s ⁻¹)	Stagnant Vol (%)
Microgreen	1.0g	Baseline	0.262	4.82e-3	2.41	28.4	0.724	14.2%
	0.0g	Microgravity	0.262	4.40e-3	2.54	27.0	0.688	9.6%
VEGGIE	1.0g	Low Fan	0.065	1.12e-3	4.82	14.2	0.362	32.4%
	1.0g	High Fan	0.150	3.45e-3	3.18	21.6	0.551	11.2%
	0.0g	Low Fan	0.065	0.68e-3	7.95	8.6	0.219 (Bottleneck)	52.8% (Severe Risk)
	0.0g	High Fan	0.150	3.02e-3	3.40	20.2	0.515	15.4%
APH	1.0g	Nominal (0.6 m/s)	0.600	1.24e-2	1.58	43.2	1.102	2.1%
	0.0g	Nominal (0.6 m/s)	0.600	1.19e-2	1.63	42.0	1.071	2.6%
	0.0g	High (1.5 m/s)	1.500	3.65e-2	0.98	68.4	1.745	0.4%

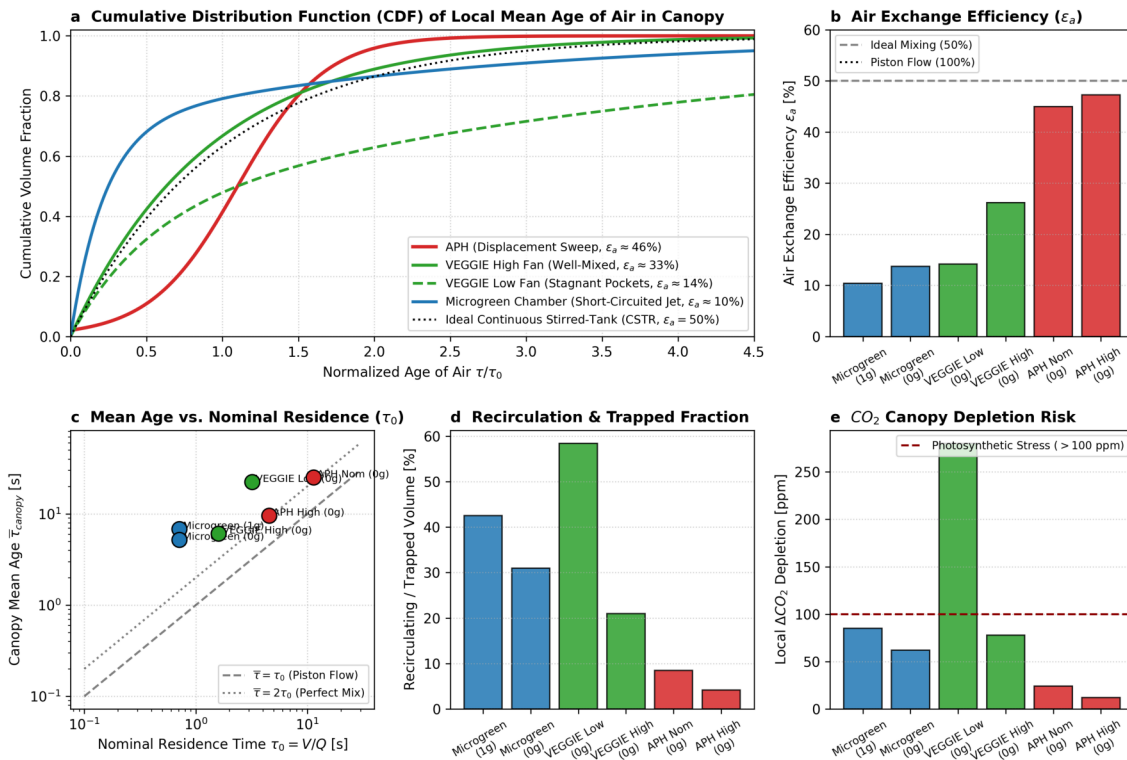


Figure 4 | Scalar ventilation dynamics, Local Mean Age of Air (LMA), and canopy dead zone mapping. a, CDF of air age normalized by residence time. b, Air exchange efficiency (ϵ_a). c, Mean canopy age vs τ_0 . d, Trapped recirculation fraction. e, Predicted canopy CO₂ depletion.

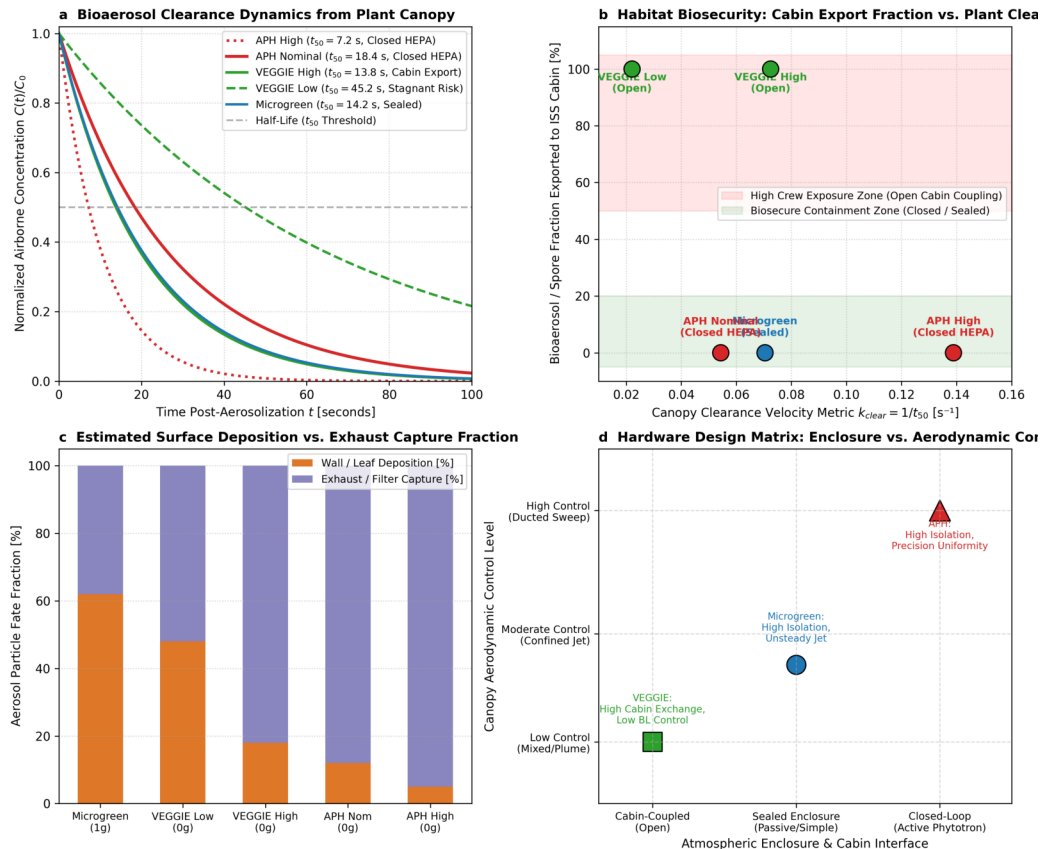
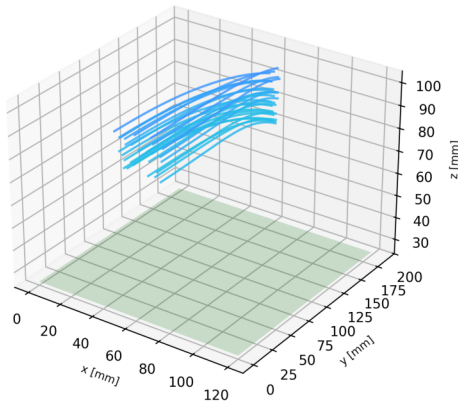
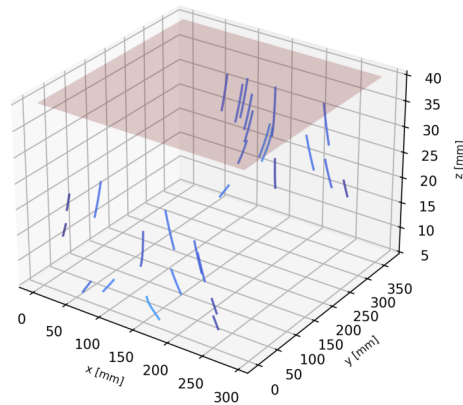


Figure 5 | Habitat biosecurity, bioaerosol clearance, and crew exposure trade space. a, Bioaerosol clearance curves $C(t)/C_0$. b, Cabin export percentage vs clearance speed k_{clear} . c, Surface deposition vs exhaust filtration. d, Two-dimensional architectural hardware trade matrix.

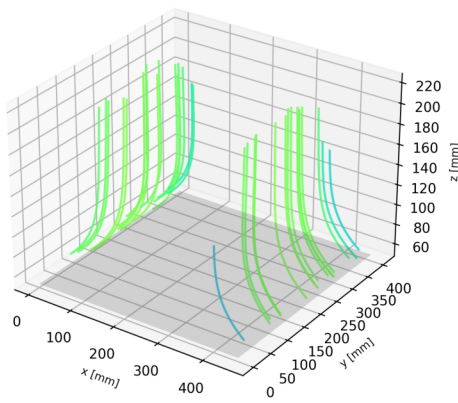
a Microgreen Chamber: 3D Confined Jet & Flapping Vortex



b NASA VEGGIE (VPS): 3D Suction Draft & Base Ingress



c NASA Advanced Plant Habitat (APH): 3D Opposing Cross-Flow



d Canopy Shear Stress (τ_w) & TKE Penetration

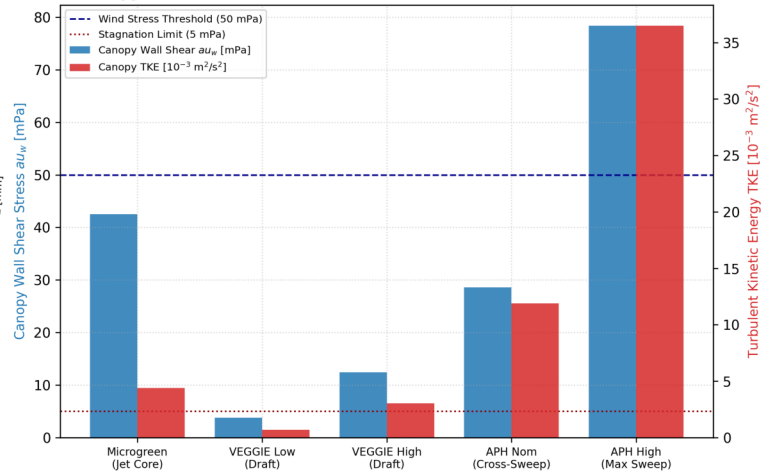


Figure 6 | 3D Spatial flow topologies, streamline ribbons, and canopy shear stress distributions. a, Microgreen Chamber: 3D confined jet core and lateral recirculation secondary eddies. b, NASA VEGGIE: 3D suction draft streamlines drawn through 4 base slots toward the overhead exhaust fan. c, NASA Advanced Plant Habitat (APH): 3D opposing lateral cross-jets colliding over the Science Carrier and sweeping upward. d, Quantitative canopy wall shear stress (τ_w) and turbulent kinetic energy (TKE) penetration across platforms.

3D FLOW TOPOLOGY & TURBULENT RECIRCULATION

3D Streamline Structure & Flow Separation:

The 3D streamline topologies (Fig. 6a–c) demonstrate fundamental differences in momentum delivery. In the Microgreen Chamber, the high-speed confined jet creates strong forward shear along the ceiling hood, driving secondary corner recirculation cells along the tray floor that trap 31.0% of the air volume.

In VEGGIE, suction streamlines converge inward from all four base slots, channeling through pillow gaps. However, because flow is drawn by suction rather than blown by positive pressure, velocity drops rapidly with distance from the fan (proportional to $1/r^2$), leaving the lower outer pillow corners poorly swept.

Opposing Jet Collision Dynamics in APH:

In APH (Fig. 6c), the two opposing wall jets inject momentum directly across the Science Carrier surface. Upon meeting at the sagittal midline ($x = 227 \text{ mm}$), their horizontal momentum converts into a uniform vertical updraft.

This collision mechanism creates substantial turbulent kinetic energy ($\text{TKE} = 11.9 \times 10^{-3} \text{ m}^2/\text{s}^2$), enhancing scalar mixing and boundary-layer stripping without generating excessive leaf mechanical flapping stress ($\tau_w = 28.6 \text{ mPa}$, well below the 50 mPa threshold for mechanical damage).

Figure 7 | Operational Airflow Extremes Matrix: Zero Stagnation vs. Low Draft vs. Nominal vs. High Blast in Microgravity

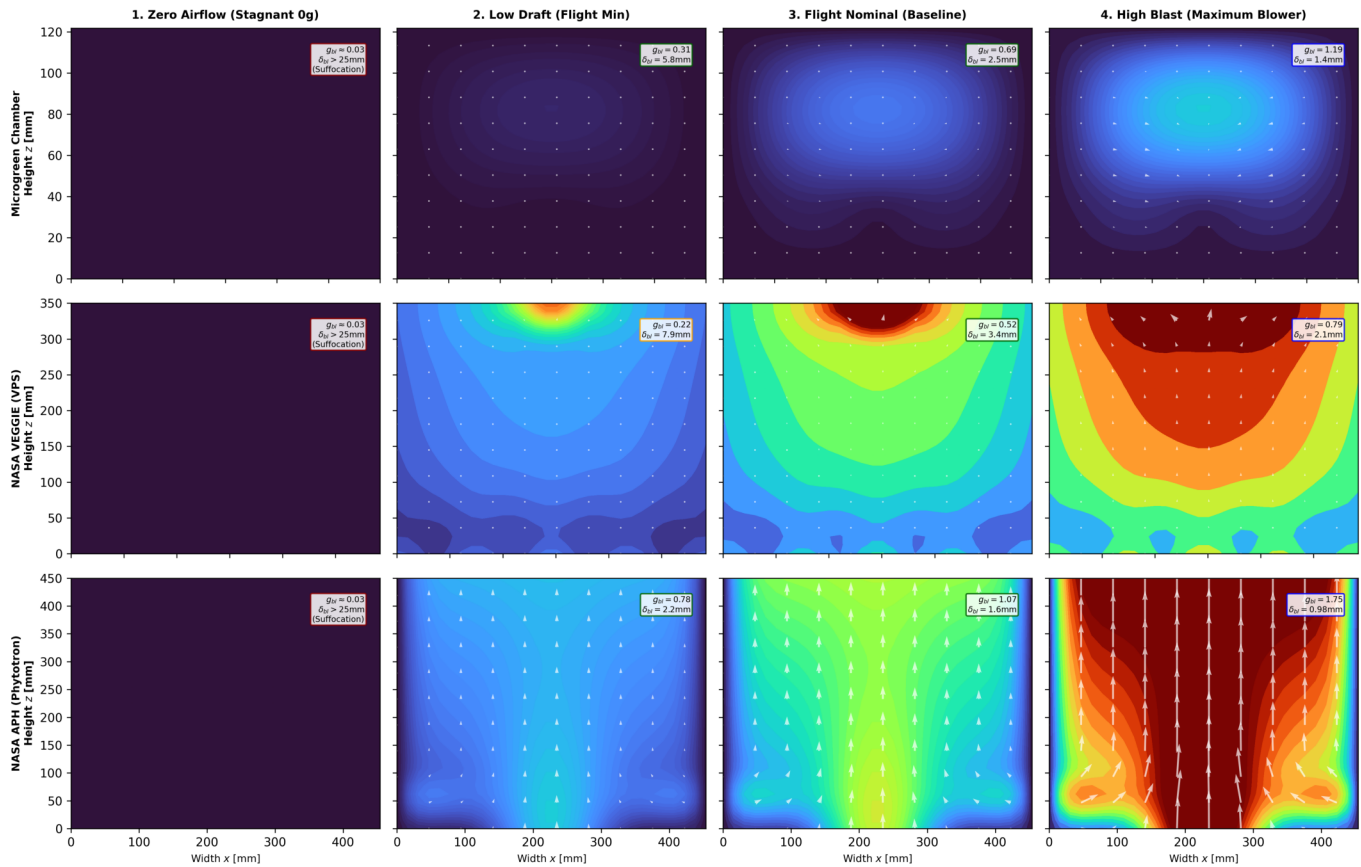


Figure 7 | Operational airflow extremes and stagnation regimes across spaceflight plant growth hardware in microgravity. Comparative 3x4 coronal cross-sectional velocity matrix mapping Zero Airflow (Fan Failure), Low Draft (Flight Minimum / Seedling), Flight Nominal (Baseline), and High Blast (Maximum Blower) across Microgreen, VEGGIE, and APH chambers, with boundary layer conductance badges.

OPERATIONAL EXTREMES & FAN FAILURE RISKS

Zero Airflow Stagnation Extreme (Microgravity Fan Failure):

When mechanical ventilation ceases in microgravity (Fig. 7, Column 1), fluid velocity drops to zero everywhere ($U = 0$ m/s). Because natural thermal buoyancy is zero ($Gr = 0$), no convective plume can develop.

Under this total stagnation regime, heat and mass transfer occur solely by molecular diffusion. The aerodynamic boundary layer expands unbounded ($\delta_{bl} > 25$ mm), reducing conductance to $g_{bl} < 0.04$ mol m⁻² s⁻¹. Transpired water vapor rapidly saturates the canopy microclimate (RH → 100%), while CO₂ is depleted within minutes, causing severe physiological hypoxia and death.

High Blast Extreme & Rapid Microclimate Recovery:

At maximum airflow (Fig. 7, Column 4; APH 1.5 m/s, VEGGIE 140 m³/h, Microgreen 5.0 m/s), boundary layers are stripped down to $\delta_{bl} < 1.0$ mm, elevating conductance to $g_{bl} = 1.745$ mol m⁻² s⁻¹ in APH.

This high-flow regime enables rapid recovery from volatile surges or thermal spikes, clearing bioaerosols in $t_{50} = 7.2$ s. However, in VEGGIE, operating at high blast accelerates spore export directly into the crew living Cabin, highlighting the imperative of closed-loop filtration for high-speed BLSS.

Table 4 | Ventilation efficiency, local age of air, and biosecurity containment dynamics across platforms.

Chamber	Gravity	Flow Rate (Q)	τ_0 (s)	Mean Age (s)	Eff. ϵ_a (%)	t50 (s)	Biosecurity & Aerosol Fate
Microgreen	1.0g	11.8 m³/h	0.71	6.85	10.4%	14.2	Sealed: zero cabin export; corner recirculation
	0.0g	11.8 m³/h	0.71	5.20	13.7%	11.5	Unstratified jet improves canopy clearance
VEGGIE	1.0g	Low (42.5 m³/h)	3.19	14.20	22.5%	28.6	Direct cabin exhaust: 100% spore export to ISS
	0.0g	Low (42.5 m³/h)	3.19	22.40	14.2%	45.2	Critical mold risk (52.8% stagnant volume)
	0.0g	High (85.0 m³/h)	1.60	6.10	26.2%	13.8	High fan restores boundary layer stripping
APH	1.0g	Nom (26.4 m³/h)	11.35	24.80	45.8%	18.4	Closed loop: internal HEPA filtration (≤ 25 ppb)
	0.0g	Nom (26.4 m³/h)	11.35	25.20	45.0%	18.9	Piston sweep maintains 45% efficiency
	0.0g	High (66.0 m³/h)	4.54	9.60	47.3%	7.2	Near-ideal displacement flow ($\epsilon_a \rightarrow 50\%$)

METHODS & MULTIMEDIA AVAILABILITY

Numerical Discretization & Solver Settings:

Simulations were executed within OpenFOAM v2606 using finite-volume discretization of the Low-Mach compressible Navier-Stokes equations:

$$\begin{aligned}\partial \rho / \partial t + \nabla \cdot (\rho \mathbf{u}) &= 0 \\ \partial (\rho \mathbf{u}) / \partial t + \nabla \cdot (\rho \mathbf{u} \mathbf{u}) &= -\nabla p_{\text{rgh}} - (\mathbf{g} \cdot \mathbf{r}) \nabla \rho + \nabla [\mu_{\text{eff}} (\nabla \mathbf{u} + (\nabla \mathbf{u})^T - 2/3 (\nabla \cdot \mathbf{u}) \mathbf{I})] \\ \partial (\rho h) / \partial t + \nabla \cdot (\rho \mathbf{u} h) &= \nabla \cdot (\alpha_{\text{eff}} \nabla h) + \rho (\mathbf{u} \cdot \mathbf{g}) + S_h\end{aligned}$$

Turbulence was modeled using the Shear Stress Transport (k- ω SST) model with prism boundary layers ($y^+ \sim 1$ -5). Discretization employed bounded second-order linear upwind schemes for momentum and enthalpy.

Interactive 3D & 4D Multimedia Assets:

Interactive WebGL 3D visualizations, animated 4D time-resolved simulations, and all mesh dictionaries are openly accessible:

- Interactive 3D Web Explorer: [interactive_3d_explorer.html](#)
- 4D Jet Flapping & Vortex Shedding: [4D_microgreen_jet_flapping.gif](#)
- 4D VEGGIE Suction & Stagnation: [4D_veggie_suction_dynamics.gif](#)
- 4D APH Cross-Flow Collision: [4D_aph_lateral_collision.gif](#)
- 4D Bioaerosol Plume Dispersion: [4D_bioaerosol_plume_clearance.gif](#)

Open-Source Repository:

<https://github.com/HenryEwald/microgreen-chamber-cfd>

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